

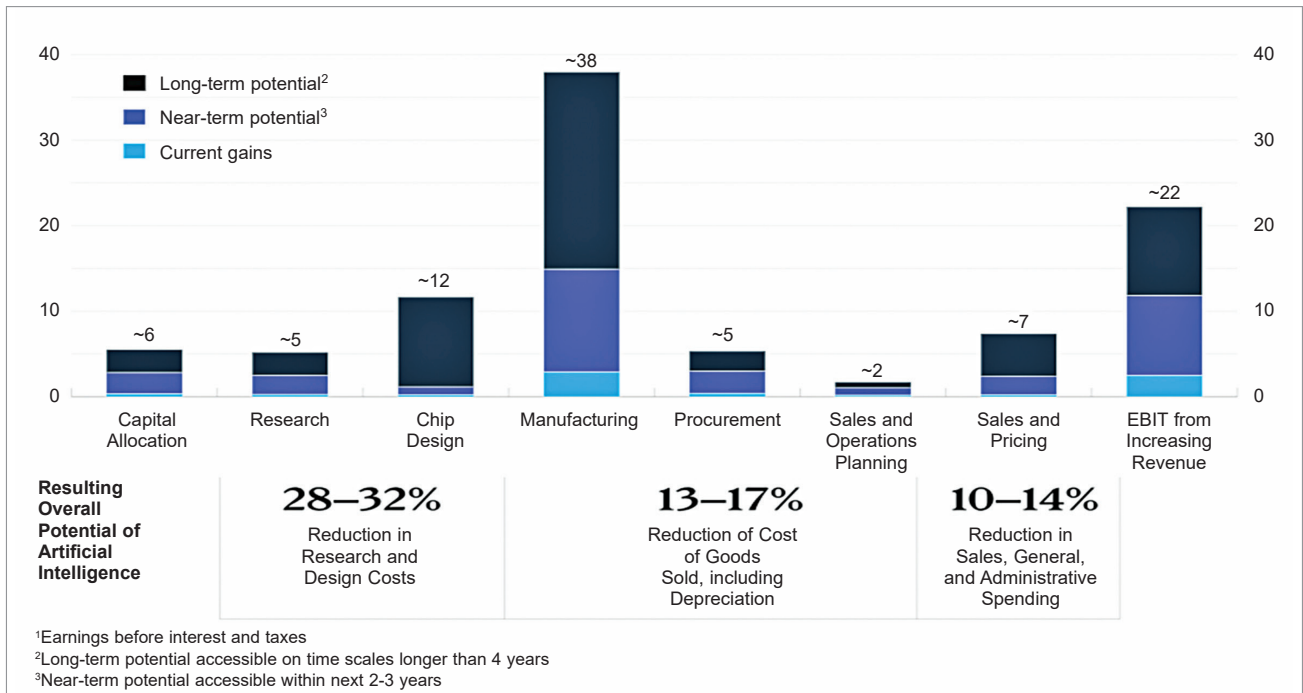


Fig. 4: Impact on earnings EBIT,¹ by semiconductor key activities, in \$ billion (Source: IBS; McKinsey Analysis)

supply chain disruptions experienced during the Covid-19 pandemic.

The semiconductor industry is on the cusp of a revolution driven by AI and customisation. As the demand for AI-powered devices continues to grow, semiconductor designers must adapt to meet the diverse needs of different applications efficiently. Software-defined hardware, domain-specific chipsets, and AI-driven automation are key trends that will shape the industry's future.

Recognising the strategic importance of AI and talent development, semiconductor companies are embracing AI as a top priority. The International Technology Roadmap for systems and devices (ITRE) is also actively contributing to the ecosystem's development. The semiconductor industry's transformation is inevitable, and those who harness the power of AI and customisation will thrive in this new era of semiconductor design and technology.

The semiconductor industry is in the midst of a transformative era,

driven by the integration of AI and the growing demand for customisation. In a recent discussion, industry experts explored the various facets of this revolution, from custom chipsets to software-defined hardware and the role of AI in reshaping semiconductor design and development. In this article, we will take a closer look at the insights and innovations shared during this enlightening conversation.

AI's role and customisation with ASICs and domain-specific chipsets

Application-specific integrated circuits (ASICs) and domain-specific chipsets, as the name suggests, are designed with a specific application or use case in mind. These chips are tailored to excel in their intended application, whether AI for autonomous vehicles or industrial automation.

What is noteworthy is that these ASICs are not one-size-fits-all solutions. Instead, they are finely tuned to meet the unique require-

ments of each application, resulting in improved efficiency and cost-effectiveness. This move towards customisation is changing the game in semiconductor design, allowing companies to offer specialised chipsets that cater to diverse industries and applications.

The software-defined hardware challenges the traditional approach of fixed hardware configurations. Instead, it introduces flexibility by enabling modules within a chip to be activated or deactivated depending on the specific application's needs.

For example, consider a Tesla car. Customers can purchase a base model and then choose to enable additional features or functionalities as needed. These features are already present in the hardware, but they can be unlocked through software activation. This approach allows for greater adaptability and efficiency in hardware usage.

AI is making its mark throughout the semiconductor lifecycle. One notable application of AI is in creating